Grading Rubric

Roll Number:
Name:

Step	Max	Awarded
(a1) k independent columns exist (definition of rank)	5	
(a2) every column is a combination of them, *with reason*	5	
(a3) arbitrary $(k+1)$-submatrix; truncation preserves combinations	10	
(a4) Fundamental Theorem on Vector Spaces: $k+1$ from $k \Rightarrow$ dependent	5	
(a5) dependent columns $\Rightarrow$ determinant 0	3	
(a6) arbitrariness of the submatrix; conclusion	2	
(b1.1) expansion along row k equals $\left\|M\left(\mathcal{A}^C\right)\right\|$	3	
(b1.2) $\left\|M\left(\mathcal{A}^C\right)\right\|$ is an order-k minor from the first k columns, $=0$	3	
(b2.1) the submatrix $M\left(A^\ell\right)$ (rows $1,\ldots,k-1,\ell$)	3	
(b2.2) $\left\|M\left(A^\ell\right)\right\| = 0$ by the standing assumption	3	
(b2.3) expansion of $\left\|M\left(A^\ell\right)\right\|$ along its last row	3	
(b2.4) its cofactors equal A_{ki} (same deleted submatrix, same sign)	5	
(b2.5) conclusion for all $j = k+1,\ldots,m$	1	
(b3.1) recognised as an alien-cofactor expansion	3	
(b3.2) vanishes (alien cofactors / repeated row)	3	
(b4.1) the three bullets cover all rows $j = 1,\ldots,m$	6	
(b4.2) vector identity $\sum_i A_{ki}A^i = \mathbf{0}$, coordinatewise	6	
(b4.3) non-trivial relation $\Rightarrow$ dependence	6	
(b5.1) independence forces $A_{k1} = \cdots = A_{kk} = 0$	3	
(b5.2) the A_{ki} are $\pm$ order-$(k-1)$ minors from those columns	3	
(b5.3) rearrangement $\Rightarrow$ *all* order-$(k-1)$ minors vanish	3	
(c1) hypothesis in coordinates, for all i, j	3	
(c2) the case $u = \mathbf{0}$	3	
(c3) $\mu = v_p/u_p$ and $v_i = \mu u_i$ for all i	7	
(c4) dependence, non-triviality, contradiction	3	
Total	**100**	